

SNR (Signal to Noise Ratio)

$$= 10 \log_{10} \frac{P_S}{P_N} \text{ dB}$$

$$\text{If } \frac{P_S}{P_N} = 10, \quad \text{SNR} = 10 \log_{10} 10 \\ = 10 \text{ dB}$$

Another way to represent Signal strength is dBm (with respect to $P_{ref} = 1 \text{ mW}$)

Ex1 if signal transmitted at 100mW

$$= 10 \log_{10} \frac{10^2}{1} \Rightarrow \underline{20 \text{ dBm}}$$

Ex2

Suppose, $P_S = 10 \text{ watt}$ $\text{SNR} = 10 \log_{10} \frac{10}{1}$
 $P_N = 1 \text{ watt}$ $= 10 \text{ dB} \checkmark$

Now, in terms of representing P_S in terms of

dBm $P_S = 10 \text{ watt} = 10 \times 10^3 \text{ mW}$

$$= 10 \log_{10} \frac{10^4 \text{ mW}}{1 \text{ mW}} = 40 \text{ dBm}$$

Noise power in dBm $P_{ref} = 1 \text{ watt} = 10^3 \text{ mW}$

$$\text{in dBm } 10 \log_{10} \frac{10^3}{1} = 30 \text{ dBm}$$

Ex 3

$$P_s = 10 \text{ watt}$$

$$P_N = 1 \text{ mW}$$

$$\text{SNR} = 10 \log_{10} \frac{10}{10^{-3}}$$

$$= 40 \text{ dB}$$

Signal power in dBm = 10 watt = 10⁴ mW

thus in dBm

$$= 10 \log_{10} \frac{10^4 \text{ mW}}{1 \text{ mW}}$$

$$= 40 \text{ dBm}$$

Noise power in dBm = 1 mW

$$= 10 \log_{10} \frac{1 \text{ mW}}{1 \text{ mW}}$$

$$= 0 \text{ dBm}$$

Ex 4:

$$P_s = 10 \text{ watt}$$

$$P_N = 1 \text{ mW}$$

$$\text{Path loss} = 30 \text{ dB}$$

Tell the SNR in dB

First SNR without path loss = $10 \log_{10} \frac{10}{10^{-3}}$

$$= 40 \text{ dB}$$

NOW, path loss = 30dB,

Hence, resultant SNR = $40 - 30$
= 10dB

$P_x = 10\text{ watt}$ convert this to dBm
= $10 \times 10^3 \text{ mW}$

$$P_N = 1 \text{ mW}$$

$$10 \log_{10} \frac{P_x}{P_N}$$

$$10 \log_{10} \frac{10^4}{1}$$

$$= 40 \text{ dBm}$$

My signal power = 100dBm convert this
to watt/mW?

$$100 = 10 \log_{10} \frac{P_x \text{ mW}}{1 \text{ mW}}$$

$$\Rightarrow P_x = 10^{10} \text{ mW}$$
$$= 10^7 \text{ watt.}$$